

## GW — lab notebook (gravitational-wave program)

K. Bhatia ORCID 0009-0007-4447-6325 repo: GW (ported from kunalb541/cosmo2, E-numbering preserved)  
Data of record: GWTC-2.1/3 PE (GWOSC/Zenodo, 76 events); GWTC-4.0 PE (Zenodo 16053484, 86 events);  
LVK TGR products (Zenodo 17461225: imr/par/rin/liv); GWTC-3 cosmology icarogw (Zenodo 5645777); PTA  
chains: NANOGrav 15yr fig-1 release, EPTA DR2 (Zenodo 8091568), PPTA DR3 (github). Every battery locked-  
preregistered before running; every headline independently re-derived; numbers of record in **results/**, per-battery  
detail in **reports/**.

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### Black-hole posterior geometry: the event atlas and its laws (E38, E40–E44)

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**E38 GW posterior-geometry atlas.** Value/shape geometry on real compact-binary posteriors: all 76 confident GWTC-1/2.1/3 events (1.54M samples; GWOSC/Zenodo PE releases), 6 black-hole planes each —  $(m_1, m_2)$ ,  $(\mathcal{M}, q)$ ,  $(q, \chi_{\text{eff}})$ ,  $(M_f, a_f)$ ,  $(d_L, \cos \iota)$ ,  $(z, M_{\text{tot}})$ ; 74/76 produce all six. The foundation for E40–E47.

**E40 GW chirp-mass lawfulness.** Is the  $(m_1, m_2)$  posterior degeneracy just the constant-chirp-mass line? Predict each event's orientation from the constant- $\mathcal{M}$  tangent at its median masses, vs a constant-total-mass null, 75 events. PASS: median  $|\Delta\psi_{\text{chirp}}| = 7.49^\circ$  vs total-mass  $20.38^\circ$ ; chirp wins 61/75; on the elongated subset (axis ratio  $\geq 3$ ,  $n=28$ )  $4.68^\circ$ ; Spearman(error, axis-ratio) =  $-0.42$  (more elongated  $\rightarrow$  tighter, as predicted). A rare POSITIVE geometric-lawfulness result — geometry is lawful where a dominant physical combination (chirp mass) exists.

**E41 GW plane lawfulness map.** A-priori correlation-sign predictions for all 6 planes, derived by 12 physicist/verifier agents blind to the data, tested against measured within-event  $\rho$  (75 events). 3/6 HIT:

**E42 BH population + precision lawfulness.** Precision test: chirp-mass fractional precision  $\sigma_{\mathcal{M}}/\mathcal{M}$  vs SNR (75 events). PASS:  $r = -0.50$  ( $p=7\times 10^{-7}$ ), OLS slope  $-1.22$  (Fisher  $-1$ ), and partial correlation with  $\log \mathcal{M}$  at fixed SNR =  $+0.94$  (the cycle-count law  $N \propto \mathcal{M}^{-5/3}$ ). Posterior SIZE is lawful — completing orientation (E40) + correlation (E41) + size (E42). Population map:  $\chi_{\text{eff}}$  mean  $+0.076$ , 36% median-negative,  $m_1$  peaks near 10 and  $35 M_\odot$ , 7 IMBH-scale remnants, 1 BNS; the geometric outliers are the physically extreme events (GW190814  $q=0.11$ , GW191219  $q=0.04$  NSBH, GW200115 NSBH, GW190425 BNS).

**E43  $\chi_{\text{eff}}$  per-event negatives.** Does any event securely have  $\chi_{\text{eff}} < 0$  (dynamical-formation signature)? From the full posteriors (75 events): 0 events with  $P(\chi_{\text{eff}} < 0) > 0.90$  (GW191109 sits at exactly 0.900, 90% CI  $[-0.60, +0.12]$  spans 0), 0 with a 95% CI below 0. NULL: E42's 36% median-negative is largely measurement scatter, not per-event-secure negatives.

**E44  $\chi_{\text{eff}}$  hierarchical.** Does the POPULATION require support below 0? Gaussian fit (flat and induced-prior-reweighted, 75 events) + a positive-truncation comparison.  $\mu \approx 0.06\text{--}0.09$ ,  $\sigma \approx 0.045\text{--}0.061$ , negative fraction median  $\sim 9\%$  — but a  $\chi_{\text{eff}} \geq 0$  truncated population fits as well or better ( $\Delta \ln L = -1.0/-0.74$ ), so the negative tail is a Gaussian-tail artifact, NOT required. Reinforces E43. (Does not overturn the LVK negative tail: a simplified pipeline — Gaussian-only, approximate prior, no selection function — simply does not independently demand it.)

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### Testing general relativity with black holes (E45–E47, E55, E57–E60)

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**E45 IMR-consistency test.** LVK GWTC-3 Tests-of-GR: the inspiral- vs post-inspiral-inferred final mass/spin, deviation  $(\Delta M_f/M_f, \Delta a_f/a_f)$ , GR = (0, 0), 18 events. GR UPHELD —  $D_1$  combined deviation  $(-0.035, +0.002)$ , GR at credible level 0.671;  $D_2$  17/18 consistent (outlier GW190814 has an absent post-inspiral, a known systematic).  $D_3$  (the framework's added test): the mean-deviation directions are COHERENT (Rayleigh  $p = 0.001$ , 13/18 with positive final-spin deviation), only partly along the shared degeneracy — a coherent sub-threshold waveform systematic that per-event tests miss.

**E46 parameterized PN deviations.** Fractional PN inspiral-phase deviations  $\delta\hat{\varphi}_i$  (GR = 0), 9 events  $\times$  10 coefficients (SEOBNRv4). GR NOT violated. The cross-event sign-coherence test isolates **dchi0** (leading 0PN):

9/9 events positive, Bonferroni  $p = 0.004$ , combined  $+0.109 [0.040, 0.177]$  excluding 0 — the documented low-PN WAVEFORM SYSTEMATIC of a single-waveform analysis, not new physics. The non-coherent combined exclusions (`dchi3/51/61`, only 2–4/9 positive) are naive-product artifacts the coherence test correctly discounts. Single waveform; multi-waveform robustness (not run) would confirm the systematic origin.

**E47 ringdown no-hair test.** Black-hole spectroscopy: fractional deviation of the 221 overtone frequency/damping ( $\delta f, \delta \tau$ ) from the Kerr prediction, GR = (0, 0), 22 events. GR/Kerr UPHELD —  $D_1$  combined  $0.75\sigma$  from GR ( $Q = 0.246$ , via a robust Gaussian product);  $D_2$  22/22 consistent;  $D_3$  ISOTROPIC (Rayleigh  $p = 0.79$ ), no coherent deviation (unlike E45/E46 — the coherent systematics are inspiral-specific, absent in the overtone). A first-pass histogram-density product spuriously reported a 99.9% “no-hair violation” that contradicted  $D_2/D_3$  (an event at  $Q=0.987$  was only  $0.25\sigma$  from GR); the Mahalanobis cross-check caught the binning artifact and the Gaussian product corrects it.

**E55 multi-waveform robustness.** E40’s chirp-mass orientation across two independent waveform families (IMRPhenomXPHM vs SEOBNRv4PHM, 59 events): PASS — the law holds for both (median  $|\Delta\psi_{\text{chirp}}| 9.1^\circ/6.1^\circ$ ) and the orientations agree to  $3.3^\circ$  (elongated). Real geometry is waveform-robust; E46’s low-PN `dchi0` coherence, by contrast, is a known waveform systematic (would not survive). DATA-BLOCK (honest): a direct same-event multi-waveform test of the E45/E46 GR-test coherences is impossible with GWTC-3 (the parameterized set is FTI/SEOBNR-only; TIGER/IMRPhenom was delayed) — would need the GWTC-2 (different events).

**E57 black-hole area theorem.** Hawking’s 2nd law from GWTC-3:  $A_{\text{final}} \geq A_{\text{initial}}$  using the inspiral progenitors (initial areas) vs the ringdown remnant (final area), disjoint/ independent (Isi 2021 setup). Only 4 loud events carry a ringdown remnant. HELD: no violations, area increases 5–79%, GW170104 strong single-event confirmation 0.92, stacked  $0.92\sigma$ . Honest: crude independent-pairing + broad inspiral spin priors give GW150914  $P=0.74$ , below Isi’s 0.97 — a consistency check, not a stringent bound.

**E58 massive graviton / Lorentz-violation.** GW-propagation gravity test: modified dispersion  $E^2 = p^2 c^2 + A p^\alpha$  over 12 events  $\times$  8 orders ( $\alpha=0$  = graviton mass). GR UPHELD: no Lorentz-violation/dispersion signal in any event or order. Graviton mass (per-event, robust): best single-event  $m_g < 6.7 \times 10^{-24}$  eV/ $c^2$ , median  $1.8 \times 10^{-23}$  (LVK combined  $1.27 \times 10^{-23}$ ). Honesty catch: a naive posterior-product combined bound gave an absurd  $5.9 \times 10^{-27}$  (3 orders too tight, grid-unstable) — LVK use a flat-in- $A$  prior and the prior samples are empty in the release, so it cannot be divided out; the per-event bounds are reported and the LVK combined quoted, not a fabricated number.

**E59 ringdown mode consistency.** Does adding the 221 overtone shift the remnant ( $M_f, a_f$ ) from the 220-fundamental value? `rin.zip` Kerr\_220 vs plain Kerr\_221, 22 events. A naive test “fails” (7 events  $> 2\sigma$ , max 6.1), BUT the shift is COHERENT —  $M_f$  decreases in 22/22 (100%) and  $a_f$  in 21/22 (95%) with the overtone. A uniform all-event down-shift is the known ringdown- overtone SYSTEMATIC (start-time/modeling; the GW150914 overtone controversy), NOT a no-hair violation. The coherence lens (as in E45/E46) prevents a false violation claim; the clean no-hair result remains E47 (221 deviation params  $\sim 0$ , isotropic).

**E60 model-independent ringdown no-hair.** The strongest-in-principle ringdown test: the MEASURED 220 QNM frequency (damped-sinusoid fit, no Kerr assumption) vs the Kerr prediction  $f_{220}(M_f, a_f)$  from the INSPIRAL remnant (Berti fit) — independent data. Kerr CONSISTENT: no event deviates  $> 2\sigma$  (max  $|z|=0.85$ ), combined  $0.34\sigma$ . GW150914 clean: Kerr 236.6 vs measured 246.8 Hz (+5%). Weak (only 3 events have both a DS ringdown and an inspiral remnant; GW170814’s DS frequency is broad/poorly constrained), GW150914 carries the confirmation. Ringdown no-hair trilogy: E47 (deviation params  $\sim 0$ ) + E59 (remnant shift = systematic) + E60 (frequency matches Kerr) all uphold the Kerr nature.

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#### Deriving the mass-plane law; the curved chirp-mass law (E65, E67)

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**E65 the E40 law derived — locked Fisher-rotation REFUTED; the posterior IS the constant- $\mathcal{M}_c$  curve.** Locked hypothesis: E40’s residual pattern (heavy events fail toward constant- $M_{\text{tot}}$ ) is a Fisher-mixture rotation — inspiral cycles measure  $\mathcal{M}_c$ , the merger band measures  $M_{\text{tot}}$ , so orientation should rotate with detector-frame mass. Model: two-direction Fisher mixture, analytic aLIGO PSD, one constant  $\kappa$  fit ONLY on the 14 locked E40-prereg events, scored out-of-sample on the other 61. OUTCOME: D1 marginal pass (7.45 vs

7.72°), D2 FAIL, D3 (parameter-free sign test) FAIL decisively — measured  $\psi$  does NOT sit between the chirp and total tangents; rotation fractions are mostly NEGATIVE (light elongated events overshoot PAST the chirp tangent, away from total), a sign no rotation model can produce. What the failure revealed (post-hoc, labelled): the posterior lies along the constant- $\mathcal{M}_c$  CURVE; the principal axis of a curved arc is its chord, not the median-point tangent. Zero-free-parameter curved law (uses only the  $\mathcal{M}_c$  median +  $q$  marginal): median error 4.85 vs 7.49° (all 75), **0.81** vs 4.68° (elongated, 6×); GW190728’s 14° “violation” reproduced to 1°. On-record GWTC-4 out-of-sample prediction:  $< 2^\circ$  for elongated O4a events. 5 contract tests; transient ernno-60 read handled by retry; headlines re-derived by hand.

**E67 GWTC-4 out-of-sample: the curved chirp-mass law PASSES at 1.26°.** The E65 prediction — locked on record (commit 8d09cbf) from GWTC-1/2.1/3 data only, prereg pushed BEFORE any GWTC-4 file was opened — scored on the fully disjoint GWTC-4.0 O4a catalog (Zenodo 16053484; 86 events, all processed, zero dropped, all files h5py-validated). **D1 PASS: median  $|\psi_{\text{curve}} - \psi_{\text{meas}}| = 1.26^\circ$  on the 19 elongated events (threshold 2°)** — 18 of 19 land at 0.2–3.3° while the tangent carries systematic 4–11° errors that the curvature term removes (GW231118\_090602: 10.7  $\rightarrow$  0.5; GW231020: 8.0  $\rightarrow$  0.2; GW230529 mass-gap: 2.9  $\rightarrow$  0.3). D2 PASS: curve beats tangent on all 86 (7.53 vs 9.36°). D3: the E40 tangent law itself replicates out-of-sample (86% wins vs 81%; Spearman  $-0.40$  vs  $-0.42$ ). The E65 finding is hereby PROMOTED from post-hoc to out-of-sample confirmed: the  $(m_1, m_2)$  posterior geometry of GW events is the constant- $\mathcal{M}_c$  curve (zero free parameters — each prediction uses only the event’s own median source  $\mathcal{M}_c$  and  $q$  marginal). Headlines re-derived independently + one event recomputed end-to-end by hand (exact match).

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#### Gravitational-wave cosmology: the spectral-siren lever (E66)

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**E66 spectral-siren H0 obstruction — REFUTED at GWTC-3 precision.** Radical claim (E61’s obstruction logic on GW): the spectral-siren H0 program is blocked by the  $\sim 33 M_\odot$  mass-peak location  $\mu_g$  acting as an uncalibrated ruler. Test: linear-response couplings on the REAL LVK GWTC-3 joint cosmology+population posterior (icarogw release,  $n = 13,533$ ; power-law+ peak, restricted flat- $\Lambda$ CDM). Locked stop-and-diagnose fired mid-run: the multivariate response  $C_{AB}C_{BB}^{-1}$  (partial) disagreed with the mandatory decile cross-check (marginal) — an apples-to-oranges comparison in the implementation, caught exactly as the two-estimator rule intends; fixed to the marginal degeneracy track, after which all sanity gates pass. RESULT: a  $4 M_\odot$  coherent  $\mu_g$  drift moves  $H_0$  by only  $-0.51$  km/s/Mpc ( $< 1 \Rightarrow$  REFUTED; robust  $-0.41/-0.32$  across SNR cuts) — an order of magnitude below the CMB-vs-SH0ES gap and inside the chain’s own  $\pm 3.3$  statistical error. What survives:  $\mu_g$  IS the single largest systematic lever (mass group 0.38 vs rate group 0.22 km/s/Mpc per  $1\sigma$ ), so the obstruction becomes live when spectral-siren precision reaches  $\sim 1$  km/s/Mpc (O5/XG) — then an independently pinned or  $z$ -dependent peak is mandatory. Conservative caveat: the restricted  $H_0 \in [65, 77]$  prior compresses couplings (lower bounds), but the verdict does not hinge on it.

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#### The nanohertz band: the GW background across pulsar-timing arrays (E68)

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**E68 cross-PTA GWB anatomy — pure degeneracy-sliding;  $\gamma$  marches with time span.** First value/shape battery in the nanohertz band, on real validated chains: NANOGrav 15yr (HD-correlated), EPTA DR2full (HD), PPTA DR3 (CRN), + EPTA DR2new as robustness; CPTA has no public GWB chain. D1: all pairs mutually consistent (1.0–1.8 $\sigma$ ), Gaussian-faithful (sliced-W ratios 0.98–1.01). D2 HEADLINE: the constraint geometry is UNIVERSAL ( $\rho(A, \gamma) = -0.85..-0.95$ ,  $\psi = 110\text{--}116^\circ$  — one measurement kernel) and the pairwise mean offsets lie **94.6–99.9%** along the common  $A$ - $\gamma$  degeneracy axis (mean 0.98; locked threshold 2/3)  $\Rightarrow$  classified degeneracy-sliding: everything the PTAs disagree about lives along the direction a common power-law fit slides when read over different bands. D3: the SMBHB circular prediction  $\gamma = 13/3$  is excluded  $> 2\sigma$  by NANOGrav ALONE ( $z = +3.03$ ; EPTA-full +0.29, PPTA +1.25) — the SH0ES pattern (E52): one experiment carries the narrative; no combined number fabricated (E58 lesson). POST-HOC (labelled):  $\hat{\gamma}$  is STRICTLY MONOTONIC in observing span (10.3  $\rightarrow$  24.7 yr: 2.79  $\rightarrow$  3.25  $\rightarrow$  3.87  $\rightarrow$  4.21), including EPTA’s internal DR2new-vs-DR2full split (1.89 $\sigma$ , same telescopes, offset also along the degeneracy) — the coherent signature of a spectrum that flattens at high frequency relative to a single power law (or a span-correlated systematic). Successor prediction on record: longer-span subsets of the same array should fit systematically steeper  $\gamma$  (preregisterable

at IPTA DR3 / NG 20yr). 4 contract tests; every headline independently re-derived; all probe values match the publications.

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### GW data beyond black holes: the neutron-star EOS transfer (E63)

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**E63 neutron-star mass-radius geometry.** First transfer of the value/shape framework out of cosmology, into the neutron-star equation of state, in the  $(M, R)$  plane. Real posteriors: NICER pulse-profile for the low-mass star J0030 (Riley 2019 X-PSI ST+PST; Miller 2019 2-oval/3-oval) and the high-mass star J0740 (Riley 2021; Miller 2021), plus GW170817 tidal  $\Lambda$  mapped to  $(M, R)$  via the quasi-universal C–Love relation. Column identifications and every headline independently re-derived; 5 contract tests pass. METHODOLOGICAL LESSON: in  $(M, R)$  the axes are incommensurable ( $M_\odot$  vs km), so the degeneracy ANGLE  $\psi$  is metric-dependent and meaningless (joint-whitening forces orthogonal long axes — a spurious  $\psi$ -gap=  $90^\circ$  on every pair until caught); the scale-free shape invariant is the correlation  $\rho(M, R)$ . RESULT D1: the NICER pipeline/model systematic is VALUE-dominated — every Riley-vs-Miller and spot-model difference is a central-value offset at fixed orientation (orientation-only value fraction 0.90–1.00,  $\Delta\rho \leq 0.08$ ); the sharpest is J0740, a +1.75 km radius offset (Miller 14.05 vs Riley 12.30) plus a  $3.2\times$  larger Miller ellipse. This is the OPPOSITE of the cosmology analog (the KiDS statistic-dependence, E48, was a precision/shape artifact). RESULT D3:  $\rho(M, R)$  is organized by MASS REGIME and robust across independent pipelines — low-mass J0030  $\rho \approx +0.86$  (steep M–R banana) vs high-mass J0740  $\rho \approx +0.25$  (flat top of the M–R curve), regime  $\Delta\rho = 0.61$ , consistent across pipelines (0.84/0.89/0.86; 0.29/0.21) — the NS analog of “the kernel sets orientation” (organizer = location on the mass–radius curve). D4 (exploratory, flagged): X-ray J0030 vs GW170817 at  $\sim 1.4 M_\odot$  is MIXED — both a  $\sim 1.5$  km value offset (GW prefers smaller  $R$ ,  $\sim 1.1\sigma$ ) and  $\rho: 0.86 \rightarrow 0.28$ , but the GW  $\rho$  is C–Love-mapping-induced. D5: all six posteriors are near-pure ( $\text{Tr } \rho^2 = 0.990\text{--}0.996$ ); the  $\sigma^2/8$  Chernoff repackaging (E62) persists. Honest limits carried up front: non-Gaussian (multimodal) NICER marginals, the mapping dependence of the GW point, and the QI repackaging.

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### GWTC-5/O4b: the curved chirp-mass law reconfirmed out-of-sample (E71)

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**E71 GWTC-5/O4b curved-law repeat.** Third disjoint catalog after GWTC-1/2.1/3 (train) and GWTC-4/O4a (E67). Prereg locked (commit d19a7a8) BEFORE any O4b file was opened; decision rules byte-identical to E67, only a declared structural file/group adapter added. GWTC-5.0 O4b PE (Zenodo 20276106 + 20348006): 104 per-event releases, 52.84 GB, ZERO dropped, fully disjoint (event dates 240413–250119, zero name-overlap with the 86-event O4a set, 12 events from the 2025 tail). D1 (the on-record E65/E67 prediction; elongated  $\text{axr} \geq 3$ ,  $n = 32$ ): median  $|\psi_{\text{curve}} - \psi_{\text{meas}}| = 1.22^\circ < 2^\circ$  — PASS (E67 gave  $1.26^\circ$ ). D2: curve beats tangent,  $5.41^\circ$  vs  $8.23^\circ$ . D3: the E40 tangent law replicates a third time (median  $8.23^\circ$ ; chirp-vs-total wins 88%; Spearman(err, axr) =  $-0.69$ , the strongest of the three catalogs). Zero free parameters — each prediction uses only the event’s own median source  $M_c$  and  $q$  marginal. Waveform-robust: the curve tracks the measured axis WITHIN every waveform family to  $< 0.7^\circ$ ; the per-event largest-analysis pick (5 of 104 events do carry a Mixed group; the locked rule uses it) shifts only the absolute orientation. Verification four legs: 17/17 contract tests; independent numpy re-derivation of D1/D2; one event hand-recomputed from raw HDF5 via a different linear-algebra path (match to  $0.004^\circ$ ); computed medians reproduce the release’s published PESummaryTable masses (exact for most, to  $< 0.6\%$  for a few late high-mass events), column identity confirmed catalog-wide. The mass-plane geometry law is now reproducible across THREE independent catalogs spanning the full GW history. POST-HOC (labelled): the largest curve residuals are all round posteriors ( $\text{axr} \sim 1$ ), the ill-defined-orientation regime the axr gate excludes — not law-breaking physics; the E72 outlier atlas must control for axr before reading residuals as anomalies.

**E72 GW geometry outlier atlas.** Prereg locked (0d1cebfb) BEFORE any residual-axis correlation. Where does the constant- $M_c$  curve law break? Searched the 32 elongated O4b events ( $\text{axr} \geq 3$ ; residual  $r = \text{err}_{\text{curve}}$  from the locked E71 results) against 9 pre-declared axes with Benjamini–Hochberg FDR ( $q = 0.05$ ) and partial-Spearman controlling axr; Tukey outlier fence; E59 coherence lens. RESULT — a clean PHYSICAL NULL: none of precession ( $\chi_p$ ), aligned spin ( $\chi_{\text{eff}}$ ), mass ratio, redshift, chirp/total mass, or secondary mass (NSBH/mass-gap proxy) survives FDR — the constant- $M_c$  geometry has NO source-class-dependent breakdown in O4b (strengthens claim 1). The only surviving correlate is waveform-model disagreement (a systematic;  $\rho = -0.64$ ,  $q_{\text{BH}} = 0.001$ , robust to drop-one  $\leq 0.036$ ), and its NEGATIVE sign says not-physics: the events the curve fits best ( $r \sim 0.01$ –

$0.2^\circ$ ) are those where waveforms disagree most — the curve tracks the waveform-ensemble consensus better than any single waveform tracks another (measurement optics; the E59 lesson again). One marginal Tukey outlier (GW240507,  $3.37^\circ$  vs fence  $3.23^\circ$ ; low-SNR, asymmetric  $q \sim 0.27$ ) logged as a WEAK preregisterable successor target, not a claim. Verify: 6 contract tests; the headline  $\rho$  and the Tukey fence independently re-derived (hand rank-Pearson; numpy percentile). Limits:  $N = 32$  low power; loudness axis data-limited (6/32 picked groups carry a network-SNR column); eccentricity and higher-mode axes absent from the released PE (out of scope).

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**The loudest event: GW250114 no-hair deep-dive (E74)**

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**E74 GW250114 deep-dive (loudest event).** A single-event characterization + synthesis (NOT a locked battery): the O4b TGR products are not public, so the ringdown spectroscopy and no-hair verdict are the PUBLISHED result (Suzuki et al. 2026, arXiv:2605.03576, orthonormal QNMs; LVK IMR via GWTC-5). GW250114 is the loudest GW event to date (network SNR  $\sim 80$ ), a near-EQUAL-mass BBH ( $m_1 = 33.8$ ,  $m_2 = 32.2 M_\odot$ ,  $q \sim 0.96$ , low spin), remnant  $M_f^{\text{src}} = 62.8 M_\odot$ ,  $a_f = 0.679$  — my PE extraction reproduces the published IMR remnant exactly (the data gate). INDEPENDENT contribution: propagating the remnant posterior through the Berti–Cardoso–Will Kerr fits gives the no-hair QNM targets —  $(2, 2, 0)$  at 248.9 Hz /  $\tau = 4.10$  ms and the  $(2, 2, 1)$  overtone at 243.4 Hz /  $\tau = 1.35$  ms (tight to  $\sim 1$  Hz, the remnant being measured to  $\sim 1.5\%$ ). The published analysis detects the  $(2, 2, 1)$  overtone at 99.9% (up from 82.5% non-orthogonal) and finds  $\delta f_{221}, \delta \gamma_{221}$  consistent with zero — Kerr no-hair UPHELD at record SNR. GW250114 is essentially a louder GW150914-twin: near-identical remnant ( $M_f \sim 68 M_\odot$  detector,  $a_f \sim 0.68$ ; GW150914-like  $f_{220} = 250$  Hz) but  $\sim 3\times$  the SNR, which is why the once-contested overtone is now a clean detection. Curved-law N/A (near-equal-mass  $\Rightarrow$  round posterior,  $\text{axr} \sim 2$ , outside the E71 elongated set). Verify: 4 QNM contract tests (GW150914 reference,  $f \propto 1/M$ , spin trend, overtone damps faster); published-value validation of all four remnant/mass quantities. Limit: the strain spectroscopy is not reproduced — the no-hair verdict is the cited published result; the area theorem awaits the O4b TGR release.